OpenFlow & SDN Concepts: Theory

1. Introduction to OpenFlow

OpenFlow is a protocol that enables the communication between the control plane (centralized controller) and the data plane (network devices like switches). It is a foundational technology for Software Defined Networking (SDN) and allows direct programming of the forwarding plane by the controller. OpenFlow standardizes how the controller interacts with network devices to dynamically manage traffic flows.

2. History and Evolution

OpenFlow was introduced around 2008 by Stanford University as a way to separate network control and forwarding functions. Before SDN, networking devices had integrated control and data planes, making management complex. OpenFlow evolved through multiple versions adding features like group tables, IPv6 support, and enhanced security.

3. Control and Data Plane Separation

In traditional networks, control plane (routing decisions) and data plane (packet forwarding) are tightly integrated in devices. SDN separates these, placing the control plane in a centralized controller and leaving the data plane in simple forwarding devices. This separation allows centralized management, easier automation, and faster innovation.

4. OpenDaylight Architecture Overview

OpenDaylight (ODL) is an open-source SDN controller platform that supports OpenFlow and other southbound protocols. Its architecture consists of:

- **Southbound Plugins:** Interfaces to network devices (e.g., OpenFlow, NETCONF).
- **Controller Core:** Manages network topology, device management, and event handling.
- **Northbound APIs:** Expose network services to applications and orchestration tools.
- **Applications:** Network services such as path computation, firewall, and load balancing.